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Probabilistic Artificial Intelligence
Questions Pack

February 7, 2024

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. You must provide your answers to the questions on the accompanying **answer sheet** by blackening out the corresponding squares. Please carefully read the following instructions regarding the answer sheet:

- During the exam, you may use pencil and eraser to mark and edit your answers on the answer sheet. After the time is up, we will collect the questions pack. Make sure that you marked all your answers by then, at least with a pencil. After collecting the questions pack, we provide you with additional **10** minutes of time to blacken out the squares on the answer sheet with a **black pen**.
- As the answer sheet will be graded by a computer, please **make sure to blacken out the whole square. Do not use ticks or crosses**.
- Nothing written on the pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.
- Please make sure that your answer sheet is clean and do not write anything on the answer sheet except the squares you blackened out. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, questions award **1, 2, 3, or 4** points if answered correctly, depending on the difficulty of the question. You will encounter the following question types:

- **True or False.**
- **Multiple choice questions with a *single* correct answer.**

  These multiple choice questions have **exactly one** correct choice.
- **Multiple choice questions with *multiple* correct answers.**

  These multiple choice questions are marked with a "♣" and have **one or more** correct choices (at least one choice is correct). You will receive full points if you have chosen **all** the correct choices and **none** of the wrong choices. You will get zero points otherwise.

There are no negative points in this exam, so **0 points** are awarded if a question is answered wrongly or not attempted. The sections and questions are not ordered by difficulty. Thus, if you find a question too difficult, it may make sense to get back to it in the end.

Questions in the same section might be related. In addition, some questions require information from previous parts. Hence, if you intend to skip some questions, make sure that you read all notes and details between the questions.

The notation $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian random distribution with mean μ and variance σ^2. For example, $\mathcal{N}(3, 4)$ designates the Gaussian distribution with mean 3 and standard deviation 2.

Good luck!

1 Bayesian Linear Regression (13 points)

1.1 Hyperpriors (10 points)

We consider a dataset $\{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$ of size n, where $\boldsymbol{x}_i \in \mathbb{R}^d$ denotes the feature vector and $y_i \in \mathbb{R}$ denotes the label of the i-th data point. Let ϵ_i be i.i.d. samples from the Gaussian distribution $\mathcal{N}(0, \lambda^{-1})$ for a given $\lambda > 0$. We collect the labels in a vector $\boldsymbol{y} \in \mathbb{R}^n$, the features in a matrix $\boldsymbol{X} \in \mathbb{R}^{n \times d}$, and the noise in a vector $\boldsymbol{\epsilon} \in \mathbb{R}^n$. The labels are generated according to

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w} + \boldsymbol{\epsilon}.$$

To perform Bayesian Linear Regression, we consider the prior distribution over the parameter vector $\boldsymbol{w}$ to be $\mathcal{N}(\boldsymbol{\mu}, \lambda^{-1}\boldsymbol{I}_d)$, where $\boldsymbol{I}_d$ denotes the d-dimensional identity matrix and $\boldsymbol{\mu} \in \mathbb{R}^d$ is a hyperparameter.

Question 1 *(2 points)*
Given this Bayesian data model, what is the conditional covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \mathrm{Var}(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{\mu}, \lambda)$?

A $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}\boldsymbol{I}_n$

B $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}(\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{I}_n)$

C $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{I}_n$

D $\boldsymbol{\Sigma}_{\boldsymbol{y}} = \lambda^{-1}(\frac{1}{n}\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{I}_n)$

Question 2 *(3 points)*
Calculate the maximum likelihood estimate $\widehat{\boldsymbol{\mu}}_1$ of the hyperparameter $\boldsymbol{\mu}$.

A $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

B $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}$

C $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}$

D $\widehat{\boldsymbol{\mu}}_1 = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}$

Question 3 *(4 points)* Since we are unsure about the hyperparameter $\boldsymbol{\mu}$, we decide to model our uncertainty about $\boldsymbol{\mu}$ by placing the *hyperprior* $\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_d)$. Is the posterior distribution $p(\boldsymbol{\mu} \mid \boldsymbol{X}, \boldsymbol{y}, \lambda)$ a Gaussian distribution? If yes, what are its mean vector $\widehat{\boldsymbol{\mu}}_2$ and covariance matrix $\boldsymbol{\Sigma}_{\boldsymbol{\mu}}$?
Hint: For a Gaussian with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ it holds that

$$\log \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{1}{2}\boldsymbol{x}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{x} + \boldsymbol{x}^\top\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \mathrm{const.}$$

A It is not a Gaussian.

B $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$

C $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$

D $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{\Sigma}_{\boldsymbol{y}}^{-1}\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$

E $\widehat{\boldsymbol{\mu}}_2 = \boldsymbol{\Sigma}_{\boldsymbol{\mu}}\boldsymbol{X}^\top\boldsymbol{y}, \quad \boldsymbol{\Sigma}_{\boldsymbol{\mu}} = (\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}$

Question 4 *(1 point)*
We now want to make a prediction at a new input $\boldsymbol{x}^* \in \mathbb{R}^d$, for which the true label is $y^* = \boldsymbol{x}^{*\top}\boldsymbol{w} + \epsilon$. The posterior variance of y^* can be shown to be

$$\mathrm{Var}\,(y^* \mid \boldsymbol{x}^*, \boldsymbol{X}, \boldsymbol{y}, \boldsymbol{\mu}, \lambda) = \lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^* + \lambda^{-1}.$$

What are the aleatoric and epistemic components of the posterior variance?

A $\lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^*$ is the aleatoric and λ^{-1} the epistemic component

B $\lambda^{-1}\boldsymbol{x}^{*\top}(\boldsymbol{X}^\top\boldsymbol{X} + \boldsymbol{I}_d)^{-1}\boldsymbol{x}^*$ is the epistemic and λ^{-1} the aleatoric component

1.2 Short Questions (3 points)

Question 5 *(1 point)*
The MAP (Maximum A Posteriori) estimate of the weights in Bayesian linear regression with a Gaussian prior and Gaussian likelihood is the same as the mean of the posterior distribution over the weights.

A True B False

Question 6 *(1 point)*
The Gaussian is the maximum entropy distribution among all distributions over $\mathbb{R}$ with a bounded mean.

A True B False

Question 7 *(1 point)*
Bayesian linear regression can be implemented *online* where the data $(\boldsymbol{x} \in \mathbb{R}^d, y \in \mathbb{R})$ is coming one-by-one in an i.i.d. stream, and the posterior distribution is updated recursively.
Can the recursive update of a single round be performed in $\mathcal{O}(d)$ time?

A True B False

2 Gaussian Processes (11 points)

For this section, we consider the data-generating process of $x_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$ given by

$$y_i = f(x_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where f is an unknown function and $\sigma > 0$ a known scalar. We model prior of the function f by a Gaussian process $f \sim \mathrm{GP}(0, k)$ with covariance function k.

2.1 Kernel Hyperparameters (4 points)

Suppose we observe the dataset $\mathcal{D} = \{(x_1, y_1), \ldots, (x_{10}, y_{10})\}$ and use the kernel

$$k(x, x') = \exp\left(-\frac{2\sin^2(\pi|x - x'|/p)}{l^2}\right) \exp\left(-\frac{(x - x')^2}{2l^2}\right),$$

where the length scale l and period length p are hyperparameters. Figure 1 shows the dataset $\mathcal{D}$ as well as the mean and 95% confidence region of the posterior $p(y|x, \mathcal{D}, l, p, \sigma^2)$ for four hyperparameter configurations:

- $l = 1, p = 0.4$
- $l = 1, p = 2$
- $l = 10, p = 0.4$
- $l = 10, p = 2$

Please match each of the plots in Figure 1 to the correct hyperparameter combination.

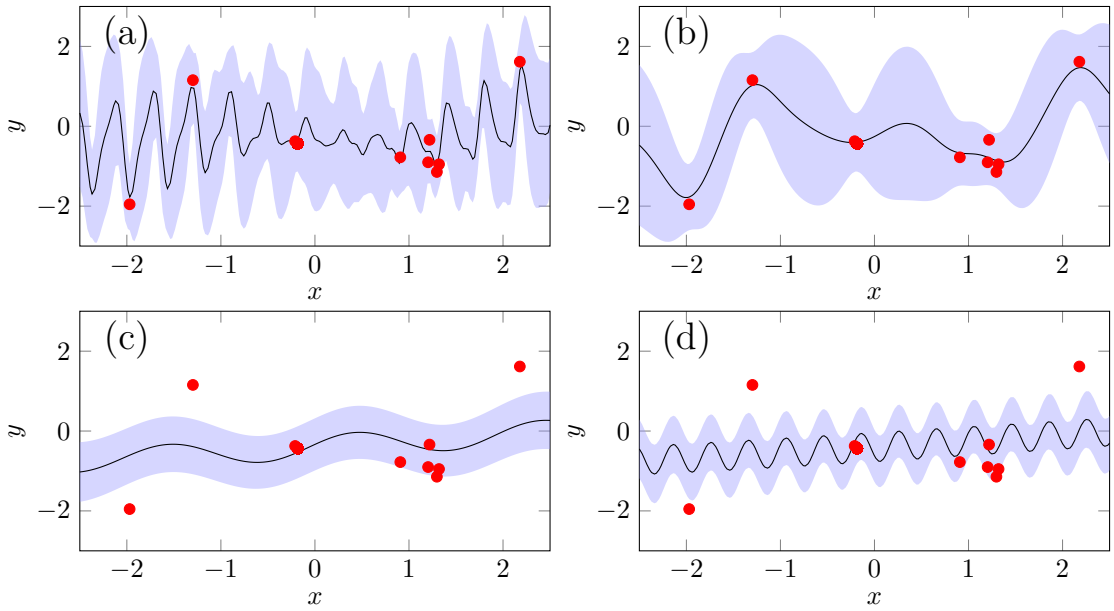


Figure 1

Question 8 *(1 points)* Which plot in Figure 1 corresponds to $l = 1, p = 0.4$?

A (a) B (b) C (c) D (d)

Question 9 *(1 points)* Which plot in Figure 1 corresponds to $l = 1, p = 2$?

A (a) B (b) C (c) D (d)

Question 10 *(1 points)* Which plot in Figure 1 corresponds to $l = 10$, $p = 0.4$?

A (a) B (b) C (c) D (d)

Question 11 *(1 points)* Which plot in Figure 1 corresponds to $l = 10$, $p = 2$?

A (a) B (b) C (c) D (d)

2.2 Model Selection (4 points)

Question 12 *(4 points)*
Suppose we observe two data points $(x_1, y_1) = (0, 1)$ and $(x_2, y_2) = (1, 0)$ in the setting described at the top of this section. We now use the linear kernel

$$k(x, x') = xx'$$

to model the unknown function f. To tune the variance hyperparameter $\sigma^2 > 0$, we maximize the marginal likelihood $p(y_1, y_2|x_1, x_2, \sigma^2)$. Which of the following values for σ^2 maximizes $\log p(y_1, y_2|x_1, x_2, \sigma^2)$? You may assume that any $\sigma^2 > 0$ satisfying $\frac{d}{d\sigma^2} \log p(y_1, y_2|x_1, x_2, \sigma^2) = 0$ is a global maximum.

Hint: It holds that $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

A $\sigma^2 = \frac{1}{2}$ B $\sigma^2 = \frac{1}{\sqrt{2}}$ C $\sigma^2 = 1$ D $\sigma^2 = \sqrt{2}$ E $\sigma^2 = 2$

2.3 Covariance Functions (3 points)

Question 13 *(1 point)*
The function $\exp(k(x, x'))$ must be a valid covariance function, assuming $k(x, x')$ is a covariance function.

A True B False

Question 14 *(1 point)*
The function $g(x, x') = k(ax, x')$ must be a valid covariance function, assuming $k(x, x')$ is a covariance function and $a > 0$ is a constant.

A True B False

Question 15 *(1 point)*
The function $g(x, x') = k(x, x') - \frac{k(x,0)k(0,x')}{k(0,0)}$ must be a valid covariance function, assuming $k(x, x')$ is a covariance function and $k(x, x) > 0$ for all $x \in \mathbb{R}$.

A True B False

3 Variational Inference and MCMC (17 Points)

3.1 Variational Inference (10 Points)

Question 16 *(1 point)*
Given a family of distributions $\mathcal{Q}$, the distribution $q^* \in \mathcal{Q}$ that minimizes $\mathrm{KL}(q \,\|\, p)$ for a distribution p is the same as the distribution that maximizes the ELBO for p.

A True B False

Question 17 *(2 points)*
We are given a dataset $\mathcal{D} = \{x_i\}_{i=1}^n$ sampled i.i.d. from a distribution $p(x \mid \theta)$ with unknown parameter θ. Assuming the prior $p(\theta)$, our goal is to approximate the posterior $p(\theta \mid \mathcal{D})$ of the unknown parameter with a distribution $q(\theta) \in \mathcal{Q}$, where $\mathcal{Q}$ is the set of all Gaussian distributions. Which of the following is necessarily true?

A If instead of maximizing the ELBO, we solve $\arg\max_{q \in \mathcal{Q}} \mathbb{E}_{\theta \sim q}[\log p(\mathcal{D} \mid \theta)]$, then the resulting distribution is the same as the distribution obtained via the Laplace approximation.

B If instead of maximizing the ELBO, we solve $\arg\max_{q \in \mathcal{Q}} \mathbb{E}_{\theta \sim q}[\log p(\mathcal{D} \mid \theta)]$, then the resulting distribution is a Dirac delta at the maximum likelihood estimator for θ.

C The ELBO maximizer is the same as the Laplace approximation.

Question 18 *(3 points)*
Consider the family of distributions $p_\theta(x) = \mathcal{N}(x; \theta, \theta)$ on $\mathbb{R}$, that is, all Gaussian distributions with mean and variance equal to $\theta > 0$. We are given a differentiable function $f : \mathbb{R} \to \mathbb{R}$ and write f' for the derivative of f. Which of the following gives the correct expression for $\frac{d}{d\theta}\mathbb{E}_{x \sim p_\theta}[f(x)]$?

A $(\theta + \sqrt{\theta})\,\mathbb{E}_{y \sim \mathcal{N}(0,1)}[f'(y)]$

B $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[f'(\sqrt{\theta}\, y + \theta)]$

C $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[f'(\theta y + \theta)]$

D $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[(1 + \frac{y}{2\sqrt{\theta}}) f'(\sqrt{\theta}\, y + \theta)]$

E $\mathbb{E}_{y \sim \mathcal{N}(0,1)}[(1 + y) f'(\theta y + \theta)]$

Question 19 *(4 points)*
We have a dataset $\{x_1, \ldots, x_n\}$ that contains i.i.d. samples from an unknown probability density p. Our goal is to train a neural network that approximates p. Let f_θ be the function defined by the neural network and consider the density $q_\theta(x) = \exp(-f_\theta(x))/Z(\theta)$, where $Z(\theta)$ is the normalization constant. Let $\{y_1, \ldots, y_m\}$ denote samples from q_θ. To approximate p, we use stochastic gradient descent to minimize the forward KL divergence $\mathrm{KL}(p \,\|\, q_\theta)$.
Which of the following is an unbiased estimator of $\nabla_\theta \mathrm{KL}(p \,\|\, q_\theta)$?

Hint: Write down the formula for the KL and compute the gradient analytically. Note that the normalization constant $Z(\theta)$ depends only on θ.

A $\frac{1}{n}\sum_{i=1}^n \exp(f_\theta(x_i)) \nabla_\theta f_\theta(x_i)$

B $\frac{1}{n}\sum_{i=1}^n \exp(f_\theta(x_i)) \nabla_\theta f_\theta(x_i) - \frac{1}{m}\sum_{j=1}^m \nabla_\theta f_\theta(y_j)$

C $\frac{1}{n}\sum_{i=1}^n \nabla_\theta f_\theta(x_i)$

D $\frac{1}{n}\sum_{i=1}^n \nabla_\theta f_\theta(x_i) - \frac{1}{m}\sum_{j=1}^m \nabla_\theta f_\theta(y_j)$

E $\frac{1}{mn}\sum_{i=1}^n \sum_{j=1}^m \exp(f_\theta(y_j)) \nabla_\theta f_\theta(x_i)$

3.2 MCMC and Sampling (7 Points)

Question 20 *(3 points)*
Consider a Markov chain $X_1, X_2, \ldots$ defined on a finite set $\mathcal{X}$ with transition probabilities $P(X_{k+1} = x' \mid X_k = x) = R(x' \mid x)$ for $k \geq 1$. Suppose that the distribution π is chosen in such a way that R and π satisfy the detailed balance property and assume that $X_1 \sim \pi$. Which of the following choices is correct?

A The chain is ergodic and therefore converges to the stationary distribution π.

B Regardless of the choice of the transition kernel R, the variables X_1 and X_2 are not independent.

C $P(X_1 = x_1, \ldots, X_n = x_n) = P(X_1 = x_n, X_2 = x_{n-1}, \ldots, X_n = x_1)$ for any $x_1, \ldots, x_n \in \mathcal{X}$.

D For n large enough, we have $P(X_n = x) > 0$ for all $x \in \mathcal{X}$.

Question 21 *(4 points)*
We want to sample from a truncated Gaussian distribution $\pi(x) \propto \exp(-x^2/2)\mathbb{I}(|x| < 1)$, where $\mathbb{I}(\cdot)$ is the indicator function. We use MCMC starting from the point 0. Which of the following algorithms has the correct stationary distribution π?

In each example, x denotes the current state of the chain, x' is the proposed state, and x^+ is the next state of the chain. All algorithms use the proposal distribution $R(x' \mid x) = \mathcal{N}(x'; (1-\gamma)x, 2\gamma)$, where $\gamma > 0$ is a step-size parameter.

A x^+ is x' if $x' \in (-1, 1)$ and otherwise is x.

B x^+ is x' if $x' \notin (-1, 1)$ and otherwise is x.

C x^+ is x' with probability $\min\left(1, \exp\left(\frac{-\gamma}{4}(x^2 - (x')^2)\right)\mathbb{I}(|x'| < 1)\right)$ and otherwise is x.

D x^+ is x' with probability $\min\left(1, \exp\left(\frac{\gamma}{4}(x^2 - (x')^2)\right)\right)$ and otherwise is x.

E x^+ is x' with probability $\min\left(1, \exp\left(\frac{\gamma}{4}(x^2 - (x')^2)\right)\mathbb{I}(|x'| < 1)\right)$ and otherwise is x.

4 Active Learning (12 points)

4.1 Mutual Information (4 points)

Let $p(x, y)$ be the joint probability density of the continuous random variables x and y defined on $\mathbb{R}$, respectively. Moreover, let

$$H(x|y) = -\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} p(x,y)\log\left(\frac{p(x,y)}{p(y)}\right)\,dx\,dy$$
$$I(x;y) = H(x) - H(x|y)$$

be the conditional entropy and mutual information as defined in the lecture, respectively.

Question 22 *(1 point)*
If x and y are independent, then $I(x; y) = 0$.

A True B False

Question 23 ♣ *(3 points)* Which of the following statements are true? Mark all that apply.

A If $I(x; y) = 0$, then $H(y|x) = 0$.
B If $I(x; y) = 0$, then $H(x) = H(y)$.
C If $I(x; y) = 0$, then $H(y|x) = H(x)$.
D If $I(x; y) = 0$, then $H(y|x) = H(y)$.

4.2 Active Learning via Information Gain (8 points)

Our goal is to learn a linear function $f(\boldsymbol{x})$ defined on the unit circle $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x}\|_2 = 1\}$ from observations generated by

$$y_t = f(\boldsymbol{x}_t) + \epsilon_t,$$

with $\epsilon_t \sim \mathcal{N}(0, 1)$. We perform Bayesian inference and place the prior $f \sim \mathrm{GP}(0, k)$ on the unknown function, where $k(\boldsymbol{x}, \boldsymbol{x}') = \boldsymbol{x}^\top \boldsymbol{x}'$.

For a set of inputs $S = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t\}$, we define the *information gain* of S as

$$F(S) := I(\boldsymbol{y}_{1:t}; f) = H(f) - H(f|\boldsymbol{y}_{1:t}),$$

where $H(f)$ is the entropy of f according to the GP prior and $H(f|\boldsymbol{y}_{1:t})$ is the conditional entropy after observing the outcomes $y_1, \ldots, y_t$. We define the *marginal information gain* $\Delta(\boldsymbol{x}|S)$ for a new point $\boldsymbol{x}$ as

$$\Delta(\boldsymbol{x}|S) = F(S \cup \{\boldsymbol{x}\}) - F(S).$$

Question 24 *(1 point)* Which of the following assertions is true?

A $\Delta(\boldsymbol{x}|S) \geq 0$ for all S and $\boldsymbol{x} \in \mathcal{X}$.
B $\Delta(\boldsymbol{x}|S) \leq 0$ for all S and $\boldsymbol{x} \in \mathcal{X}$.

Question 25 *(1 point)* Maximizing the differential information gain $\Delta(\boldsymbol{x}|S)$ is equivalent to picking the point $\boldsymbol{x} \in \mathcal{X}$ with maximum variance of the current posterior.

A True B False

Question 26 *(2 point)* Which of the following is true *and* expresses that the function $S \mapsto F(S)$ is monotone submodular?

A For all S and S' such that $S \subseteq S'$, $F(S) \leq F(S')$.

B For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \leq \Delta(\boldsymbol{x}|S')$, for all $\boldsymbol{x} \in \mathcal{X}$.

C For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \geq \Delta(\boldsymbol{x}|S')$, for all $\boldsymbol{x} \in \mathcal{X}$.

D For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \leq \Delta(\boldsymbol{x}'|S')$, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}$.

E For all S and S' such that $S \subseteq S'$: $\Delta(\boldsymbol{x}|S) \geq \Delta(\boldsymbol{x}'|S')$, for all $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}$.

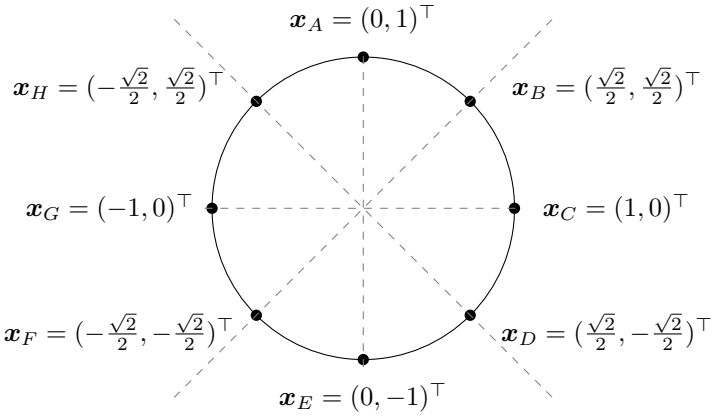


Figure 2: The unit ball in two dimensions. Points are symmetrically placed.

Question 27 ♣ *(4 points)*
Consider a greedy active learning algorithm that takes actions according to

$$\boldsymbol{x}_{t+1} \in \arg\max_{\boldsymbol{x} \in \mathcal{X}} \Delta(\boldsymbol{x}|S_t)$$

for all iterations $t \geq 1$ where $S_t = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t\}$ and recall that $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^2 : \|\boldsymbol{x}\|_2 = 1\}$. Suppose $\boldsymbol{x}_1 = \boldsymbol{x}_A = (0,1)$ as shown in Figure 2. Which of the points shown in Figure 2 are valid as the next action $\boldsymbol{x}_2$ according to the criterion of the algorithm? Mark all that apply.

Hint: Note that $F(S) = \frac{1}{2}\log\det(\boldsymbol{I} + \boldsymbol{K}_S)$ where $\boldsymbol{I}$ is the identity matrix and $\boldsymbol{K}_S$ is the kernel matrix calculated for the points in S, that is, with entries $[\boldsymbol{K}_S]_{ij} := k(\boldsymbol{x}_i, \boldsymbol{x}_j)$. Moreover, it holds that

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

A $\boldsymbol{x}_A$ B $\boldsymbol{x}_B$ C $\boldsymbol{x}_C$ D $\boldsymbol{x}_D$ E $\boldsymbol{x}_E$ F $\boldsymbol{x}_F$ G $\boldsymbol{x}_G$ H $\boldsymbol{x}_H$

5 Reinforcement Learning (18 points)

5.1 Markov Decision Processes (9 points)

Your friend Ava is a professional runner, and you want to help her by designing a daily training plan. Your aim is to maximize the cumulative running form across T days. On a particular day, *running form* is the function $r(\alpha, \beta) = 1 + \beta - \alpha$ of the *fatigue* $\alpha \in \mathbb{R}_{\geq 0}$ and the *fitness* $\beta \in \mathbb{R}$. During each day, Ava can either *train* or *recover*. Your task is to design a policy that tells her what to do on each day t depending on her fatigue α_t and fitness β_t. Ava's initial fatigue is $\alpha_0 = 1$ and her initial fitness is $\beta_0 = 0$. We assume that fatigue and fitness follow the following deterministic dynamics:

$$\log \alpha_{t+1} = \begin{cases} \log \alpha_t + 1 & \text{when training on day } t \\ \log \alpha_t - 1 & \text{when recovering on day } t \end{cases}, \quad \beta_{t+1} = \begin{cases} \beta_t + (e-1) & \text{when training on day } t \\ \beta_t - (e-1) & \text{when recovering on day } t \end{cases}$$

Throughout this problem, you may assume for calculations that $e = 2.7$.

Question 28 *(1 point)*
Do the sequences of α_t and β_t (i.e., the dynamics of fatigue and fitness) satisfy the Markov assumption?

A True B False

Question 29 *(2 points)*
Let $R_t = r(\alpha_t, \beta_t)$ be the running form on day t. A_t are the number of training days and B_t are the number of recovery days up to (not including) day t. The running form can be expressed as the following function of A_t and B_t only:

$$R_t = 1 + (e-1)(A_t - B_t) - e^{A_t - B_t}.$$

What is the maximum value for the running form on day t?

A 0
B 5
C 4
D The running form is unbounded

Question 30 *(3 points)*
The value of a policy π until day T is given by

$$J_T(\pi) = \mathbb{E}\left[\sum_{t=0}^{T-1} R_t\right].$$

Consider the policy π_1 according to which Ava trains each day. What is $J_3(\pi_1)$?

A $J_3(\pi_1) \approx -1$
B $J_3(\pi_1) \approx 0$
C $J_3(\pi_1) \approx -3$
D $J_3(\pi_1) \approx -4$

Question 31 *(3 points)*
The class of all Markovian policies is $\Pi = \{\pi(\alpha_t, \beta_t) \in \{\text{train}, \text{recover}\}\}$ where $\pi(\alpha_t, \beta_t)$ determines whether to "train" or "recover" on day $t+1$. We say that a policy π is *optimal* if

$$\pi \in \arg\max_{\pi \in \Pi} J_3(\pi).$$

Consider the following family of policies $\widehat{\Pi} = \{\pi_p : p \in [0,1]\} \subseteq \Pi$ where π_p makes a decision independently of α_t, β_t:

$$\pi_p \equiv \begin{cases} \text{train} & \text{with probability } p \\ \text{recover} & \text{with probability } 1-p. \end{cases}$$

For example, according to the policy $\pi_{0.5}$, Ava would train or recover with probability $\frac{1}{2}$ each on every day. You now wonder which policy is optimal according to the definition of optimality given above. Which of the following statements is true?

A Policy π_1 is optimal.

B Policy $\pi_{0.5}$ is optimal.

C Policies $\pi_1, \pi_{0.5}$ are suboptimal, but there exists an optimal policy in $\widehat{\Pi}$.

D $\widehat{\Pi}$ does not contain an optimal policy.

5.2 Policy Gradient and Actor-Critic Methods (5 points)

Policy gradient methods learn a parameterized policy $\pi_{\boldsymbol{\theta}}$ by gradient ascent on the expected return

$$J(\boldsymbol{\theta}) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t\right].$$

The expectations in this subsection are always taken over trajectories realized under the policy π_θ. To perform gradient ascent, one needs to obtain an estimate of the gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$. In the lecture, you were introduced to the *policy gradient theorem*, which uses the *score function trick* to derive that

$$\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \mathbb{E}[Q(x,a) \nabla_{\boldsymbol{\theta}} \log \pi_{\boldsymbol{\theta}}(a \mid x)]$$

where Q is the state-action value function of policy $\pi_{\boldsymbol{\theta}}$.

Question 32 *(1 point)*
For MDPs with a large state space and small action space, such as the MDP that models running form which was described in the previous section, policy gradient methods such as REINFORCE are computationally better suited than approximate Q-learning (such as DQN).

A True B False

Question 33 *(2 points)*
Let $a \in \mathbb{R}$ and consider the stochastic policy

$$\pi_{\boldsymbol{\theta}}(a \mid x) = \mathcal{N}(a; f_{\boldsymbol{\theta}}(x), \sigma^2).$$

Starting from the policy gradient theorem, which expression does the policy gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$ reduce to?

A $\mathbb{E}[Q(x,a)(f_{\boldsymbol{\theta}}(x) - a)\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)]$

B $\mathbb{E}[Q(x,a)(a - f_{\boldsymbol{\theta}}(x))\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)]$

C $\mathbb{E}\left[Q(x,a)\frac{(a - f_{\boldsymbol{\theta}}(x))}{2\sigma^2}\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)\right]$

D $\mathbb{E}\left[Q(x,a)\frac{(a - f_{\boldsymbol{\theta}}(x))}{\sigma^2}\nabla_{\boldsymbol{\theta}} f_{\boldsymbol{\theta}}(x)\right]$

Question 34 *(2 points)*
For the Gaussian policy from the previous question, we can use the reparameterization rather than the score function trick to obtain an unbiased estimate of the policy gradient $\nabla_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$.

A True B False

5.3 Short Questions (4 points)

Question 35 *(1 point)*
The multi-armed bandit problem is a special case of the reinforcement learning problem.

A True B False

Question 36 *(1 point)*
An off-policy method can estimate the value function of a policy π by using data collected by following a different policy π'.

A True B False

Question 37 *(1 point)*
Recall that every MDP with finitely many states and actions possesses a deterministic optimal policy. Entropy regularization is used in reinforcement learning to penalize policies with high uncertainty, acting as a bias towards deterministic policies.

A True B False

Question 38 *(1 point)*
Double DQN addresses the maximization bias of DQN (Deep Q-Network) by using in its loss $L(\boldsymbol{\theta})$ the target

$$r + \gamma Q(x', \arg\max_a Q(x', a; \boldsymbol{\theta}^{\text{old}}); \boldsymbol{\theta}^{\text{old}})$$

where the action a is selected with respect to the previous Q-network, rather than the target

$$r + \gamma Q(x', \arg\max_a Q(x', a; \boldsymbol{\theta}); \boldsymbol{\theta}^{\text{old}})$$

where a is selected with respect to the current Q-network.

A True B False





+1/13/47+

Answer Sheet of the Probabilistic Artificial Intelligence 2023/24 Exam

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

⟵ Please encode your student number on the left, and write your first and last names below.

Firstname and Lastname:

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Question 1: A B C D
Question 2: A B C D
Question 3: A B C D E
Question 4: A B
Question 5: A B
Question 6: A B
Question 7: A B
Question 8: A B C D
Question 9: A B C D
Question 10: A B C D
Question 11: A B C D
Question 12: A B C D E
Question 13: A B
Question 14: A B
Question 15: A B
Question 16: A B
Question 17: A B C
Question 18: A B C D E
Question 19: A B C D E
Question 20: A B C D
Question 21: A B C D E
Question 22: A B
Question 23: A B C D
Question 24: A B
Question 25: A B
Question 26: A B C D E
Question 27: A B C D E F G H
Question 28: A B
Question 29: A B C D
Question 30: A B C D
Question 31: A B C D
Question 32: A B
Question 33: A B C D
Question 34: A B
Question 35: A B
Question 36: A B
Question 37: A B
Question 38: A B